

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Barring tool, Part Number 3824591, or equivalent
- Feeler gauge, Part Number 3163530, or equivalent

Additional Service Items

- No additional service items required.

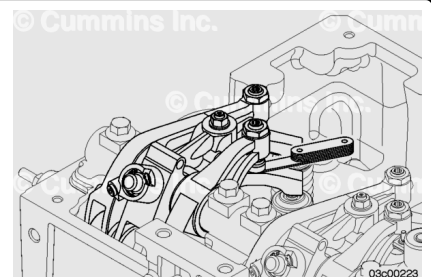
General Information

Overhead setting is **only** required at the interval specified in the appropriate Operation and Maintenance / Owners manuals, or when engine repairs cause removal of the rocker levers and/or loosening of the adjusting screws.

Excessive valve lash prior to a repair or scheduled adjustment can indicate:

- Incorrect valve set from a previous repair.
- Worn valve stems, crossheads, push rods, and/or rocker levers.

Loose rocker lever lash and the need to frequently reset the valve clearance can also indicate camshaft lobe wear or tappet wear. If an inspection of the rocker levers, valve stems, crossheads, and push rods do **not** show wear, tappet and/or camshaft lobe wear can be suspected.



Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. See equipment manufacturer service information.
- Remove the EGR crossover tube. Refer to Procedure 011-070 in Section 11.
(/qs3/pubsys2/xml/en/procedures/377/377-011-070-tr.html)
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3.
(/qs3/pubsys2/xml/en/procedures/377/377-003-011-tr.html)

Make sure the engine base timing is properly set before attempting to adjust/set the overhead. Refer to Procedure 001-088 in Section 1. (/qs3/pubsys2/xml/en/procedures/377/377-001-088.html)

All overhead valve and brake adjustments **must** be made when the engine is cold (any stabilized coolant temperature at 60°C [140°F] or below).

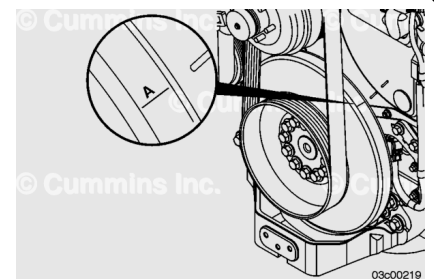
Measure

Locate the valve set marks on the outside of the vibration damper and the alignment mark on the gear cover.

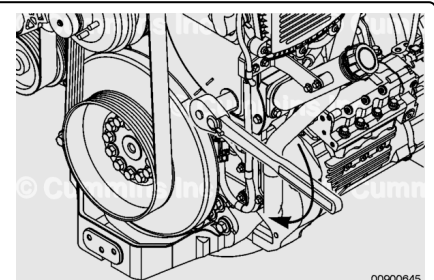
The set marks are A, B, and C:

- Set to mark A to adjust cylinder 1 or 6.
- Set to mark B to adjust cylinder 2 or 5.
- Set to mark C to adjust cylinder 3 or 4.

Two complete revolutions are required to set all valves and engine brakes.



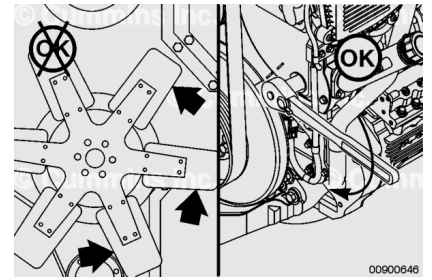
Use the following procedure for instructions on barring the engine. Refer to Procedure 000-017 in Section 0.
(/qs3/pubsys2/xml/en/procedures/377/377-000-017.html)



⚠ WARNING ⚠

Do not pull or pry on the fan to manually rotate the engine. To do so can damage the fan blades. Damaged fan blades can cause premature fan failures which can result in serious

personal injury or property damage.



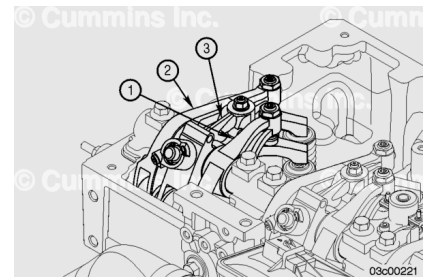
The crankshaft rotation is **clockwise**, as viewed from the front of the engine.

The cylinders are numbered from the front of the engine (1-2-3-4-5-6).

The engine firing order is 1-5-3-6-2-4.

Each cylinder has three rocker levers:

- The exhaust valve rocker lever (1)
- The intake valve rocker lever (2)
- The engine brake rocker lever, if equipped (3).



The intake valve rocker lever is **always** the long lever on the valve rocker lever shaft.

The valves and brakes on the same cylinder are adjusted at the same index mark on the vibration damper.

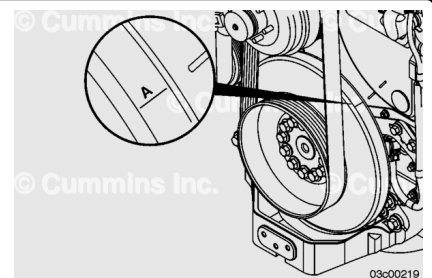
Valve and Brake Adjustment Sequence

Bar Engine in Direction of Rotation	Pulley Position	Set Cylinder Valve	Set Cylinder Brake
Start	A	1	1
Advance to	B	5	5
Advance to	C	3	3
Advance to	A	6	6
Advance to	B	2	2
Advance to	C	4	4

Firing Order: 1-5-3-6-2-4

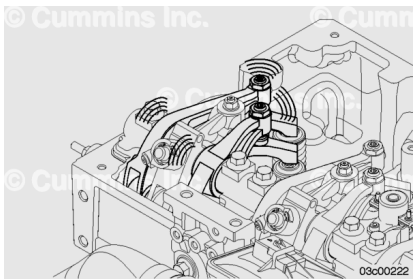
Note : For illustrative purposes, position A is shown as the first step. It is **not** necessary to start with position A, as long as the proper sequence is followed.

Use the compressor drive or barring device to bar the engine over in the direction of engine rotation, **clockwise** as viewed from the front of the engine. Align the A mark on the vibration damper with the pointer on the gear cover.

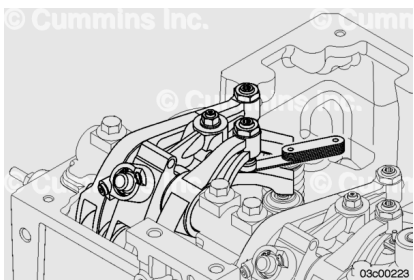


Check the valve rocker levers on the given cylinder to see if both intake and exhaust valves are closed.

Both sets of valves are closed when the rocker levers and the brake lever are loose. If both sets of valves are **not** closed, rotate the compressor drive gear one complete revolution, and align the A mark on the front damper with the pointer again.



Use feeler gauges to measure the amount of clearance (lash) between the crosshead and the rocker lever nose. Measure and record the intake, exhaust, and brake valve lash. If the valve lash is **not** within the specifications listed below, the valve **must** be adjusted. See the Adjust section in this procedure.

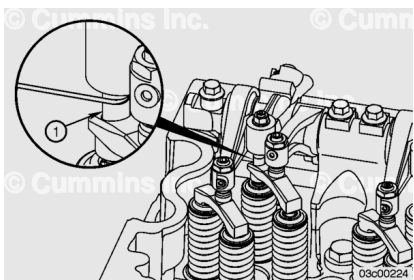


Press the brake lever down to verify the camshaft follower is in contact with the camshaft prior to measurement of the brake lash.

Valve and Brake Lash Recheck Limits

	mm		in
Intake	0.23	MIN	0.009
	0.48	MAX	0.019
Exhaust	0.56	MIN	0.022
	0.81	MAX	0.032
Brake	6.87	MIN	0.271
	7.13	MAX	0.281

Rotate the brake lever to the detent (neutral) position. Check the clearance between the engine brake actuator piston and the crosshead guide pin (1). If the brake running clearance is **not** within the specifications listed below, the running clearance **must** be adjusted. See the Adjust section in this procedure.



Brake Running Clearance

mm		in
0.635	MIN	0.025
2.79	MAX	0.110

If adjustment is **not** required, continue with the following:

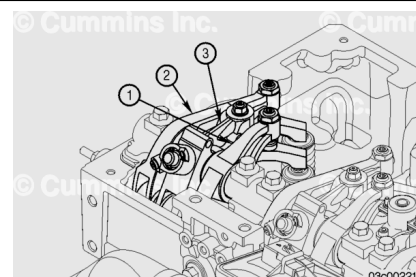
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3.
(/qs3/pubsys2/xml/en/procedures/377/377-003-011-tr.html)
- Install the EGR crossover tube. Refer to Procedure 011-070 in Section 11.
(/qs3/pubsys2/xml/en/procedures/377/377-011-070-tr.html)

Adjust

Read the entire procedure for overhead adjustment before attempting to perform this operation.

Valves and engine brakes, if equipped, **must** be correctly adjusted for the engine to operate efficiently. Valve and engine brake adjustment **must** be performed using the values listed in this section.

After an engine rebuild or any major repair where the valve setting **must** be disturbed, set all of the valves and brakes.



Valve and Brake Adjustment Values

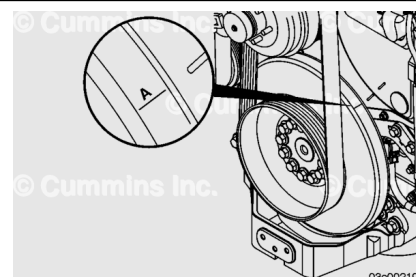
Intake Valve	0.35 mm [0.014 in]
Exhaust Valve	0.68 mm [0.027 in]
Engine Brake	7.00 mm [0.276 in]

Locate the valve set marks on the outside of the vibration damper and alignment mark on the upper gear cover.

The set marks are A, B, and C:

- Set to mark A to adjust cylinder 1 or 6.
- Set to mark B to adjust cylinder 2 or 5.
- Set to mark C to adjust cylinder 3 or 4.

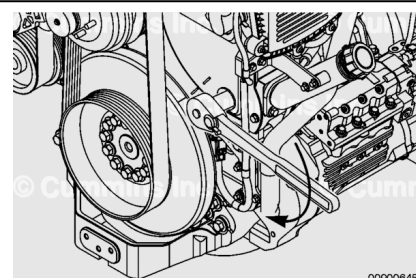
Two complete revolutions are required to set all valves, engine brakes, and injectors.



Use the following procedure for engine barring instructions.

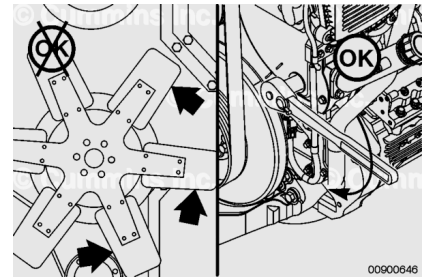
Refer to Procedure 000-017 in Section 0.

(/qs3/pubsys2/xml/en/procedures/377/377-000-017.html)



⚠ CAUTION ⚠

Do not pull or pry on the fan to manually rotate the engine. To do so can damage the fan blades. Damaged fan blades can cause premature fan malfunction which can result in serious personal injury or property damage.



The crankshaft rotation is **clockwise** as viewed from the front of the engine.

The cylinders are numbered from the front of the engine (1-2-3-4-5-6).

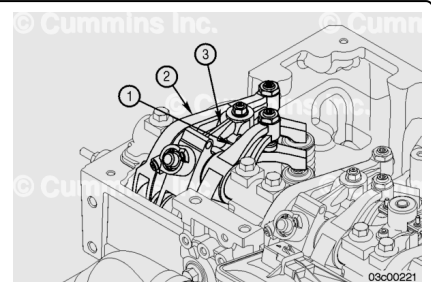
The engine firing order is 1-5-3-6-2-4.

Each cylinder has three rocker levers:

- The exhaust valve rocker lever (1)
- The intake valve rocker lever (2)
- The engine brake rocker lever, if equipped (3).

The intake valve rocker lever is **always** the long lever on the valve rocker lever shaft.

The valves and brakes on the same cylinder are adjusted at the same index mark on the vibration damper.

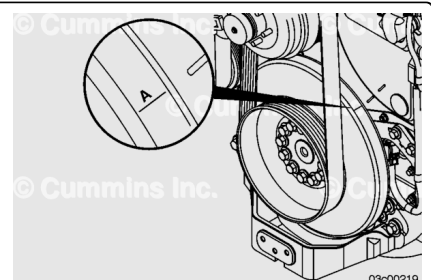


Valve and Brake Adjustment Sequence

Bar Engine in Direction of Rotation	Pulley Position	Set Cylinder Valve	Set Cylinder Brake
Start	A	1	1
Advance to	B	5	5
Advance to	C	3	3
Advance to	A	6	6
Advance to	B	2	2
Advance to	C	4	4
Firing Order: 1-5-3-6-2-4			

Note : For illustrative purposes, position A is shown as the first step. It is **not** necessary to start with position A, as long as the proper sequence is followed.

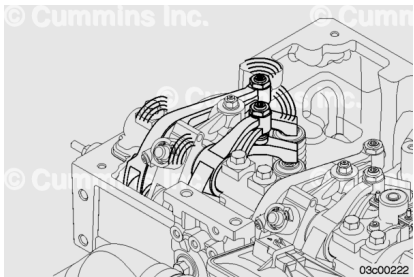
Use the compressor drive or barring device to bar the engine over in the direction of engine rotation, **clockwise** as viewed from the front of the engine. Align the A mark on the vibration



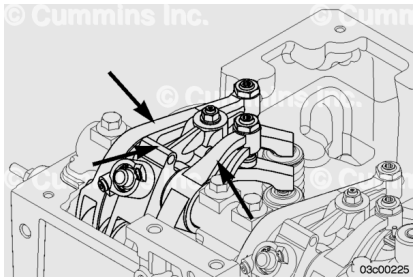
damper with the pointer on the gear cover.

Check the valve rocker levers on the given cylinder to see if both intake and exhaust valves are closed.

Both sets of valves are closed when the rocker levers and the brake lever are loose. If both sets of valves are **not** closed, rotate the compressor drive gear one complete revolution, and align the A mark on the front damper with the pointer again.

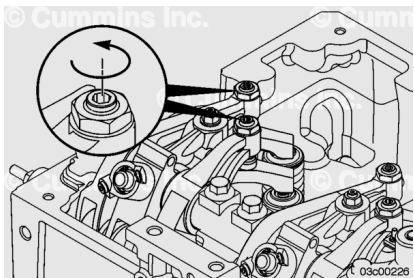


Set the valves and engine brakes on the same cylinder.



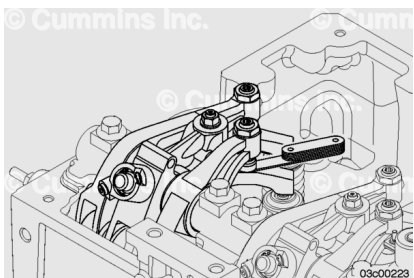
With the set mark aligned with the pointer on the gear cover and both sets of valves closed on the cylinder, loosen the locknuts on the intake and exhaust valve adjusting screws.

Back out the adjusting screws one or two turns.



Select a feeler gauge for the correct valve lash specification.

Valve Lash Specification			
	mm		in
Intake	0.36	NOM	0.014
Exhaust	0.69	NOM	0.027

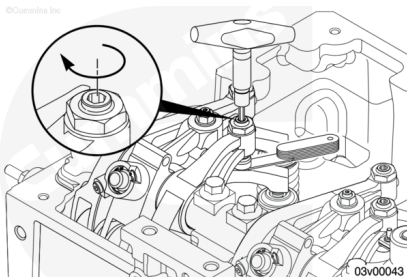


Insert the feeler gauge between the top of the crosshead and the rocker lever nose pad.

Make sure the feeler gauge is completely under the pivoting rocker lever nose pad.

Tighten the adjusting screw.

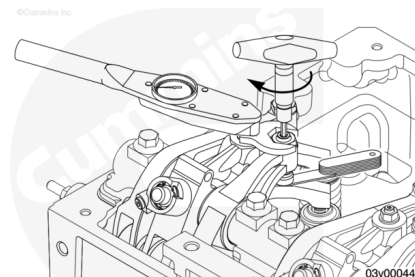
Torque Value: 0.6 n·m [5 in-lb]



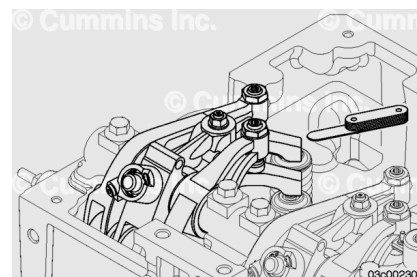
Use a torque wrench and crows foot adapter to tighten the locknut.

Hold the adjusting screw in this position. The adjusting screw **must not** turn when the locknut is tightened.

Torque Value: 45 n•m [33 ft-lb]



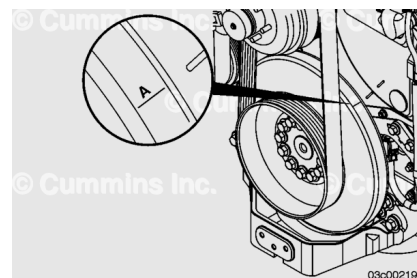
After tightening the locknut to the correct torque value, remove the feeler gauge.



⚠ CAUTION ⚠

To get maximum brake operating efficiency and to prevent engine damage, the brake adjustment instructions must be followed.

For brake adjustment, locate the valve set marks on the outside of the vibration damper and alignment mark on the upper gear cover.



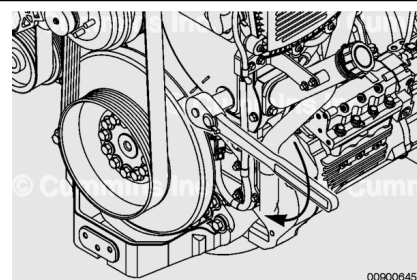
Use the following procedure for engine barring instructions.

Refer to Procedure 000-017 in Section 0.

(/qs3/pubsys2/xml/en/procedures/377/377-000-017.html)

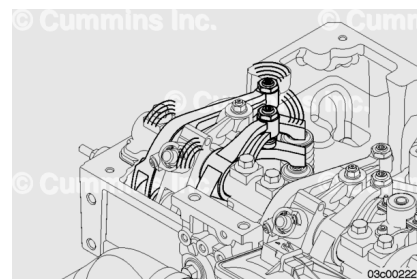
Align the A mark on the vibration damper with the pointer on the gear cover.

For illustrative purposes, position A is shown as the first step. It is **not** necessary to start with position A, as long as the proper sequence is followed.

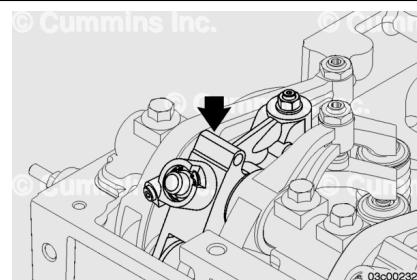


Check the valve rocker levers on the given cylinder to see if both intake and exhaust valves are closed.

Both sets of valves are closed when the rocker levers and the brake lever are loose. If both sets of valves are **not** closed, rotate the compressor drive gear one complete revolution, and align the A mark on the front damper with the pointer again.

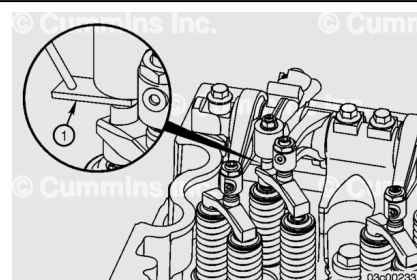


Press the engine brake lever down to verify the camshaft follower is in contact with the camshaft.



Loosen the locknut on the brake lever adjusting screw, and back out the adjusting screw one turn.

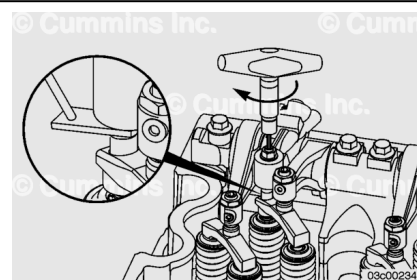
Insert the feeler gauge (1), Part Number 3163530, or equivalent, between the bottom of the engine brake piston and the top of the exhaust valve pin on the exhaust valve crosshead.



Brake Lever Lash Specification

mm		in
7.00	NOM	0.276

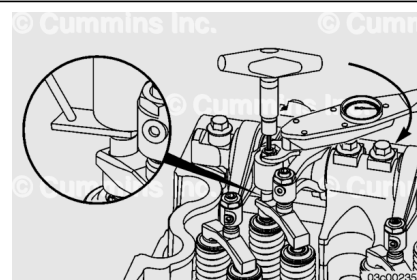
Tighten the adjusting screw until drag on the feeler gauge is felt. Proper drag means that there is no motion of the brake lever camshaft follower against the cam lobe.



Hold the engine brake lever adjusting screw and tighten the locknut.

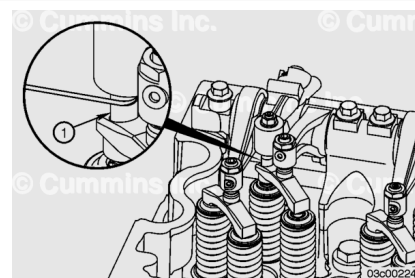
Torque Value: 20 n•m [177 in-lb]

Remove the feeler gauge.



⚠ CAUTION ⚠

Engine damage can occur if the running clearance is not within specifications.



Check the running clearance:

1. Rotate the engine brake rocker lever to the detent (neutral) position.
2. Check the clearance between the engine brake lever actuator piston and the crosshead guide pin.

Engine Brake Rocker Lever Running Clearance

mm		in
0.635	MIN	0.025
2.790	MAX	0.110

If the running clearance does **not** meet specification, loosen, but do **not** remove the rocker shaft capscrews, and rotate the shaft in the direction required to bring the running clearance within the listed specification. It is critical that this clearance be set and verified on all six brake levers. Engine damage can result if this task is **not** completed.

Check the brake running clearance.

The rocker lever shafts **must** be adjusted so that all three engine brake levers fall within the given running clearance specification.

Repeat the process to adjust all engine brakes, and valves according to the chart below.

Valve and Brake Adjustment Sequence

Bar Engine in Direction of Rotation	Pulley Position	Set Cylinder Valve	Set Cylinder Brake
Start	A	1	1
Advance to	B	5	5
Advance to	C	3	3
Advance to	A	6	6
Advance to	B	2	2
Advance to	C	4	4

Firing Order: 1-5-3-6-2-4

Finishing Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the rocker lever cover and gasket. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/377/377-003-011-tr.html>)
- Install the EGR crossover tube. Refer to Procedure 011-070 in Section 11. (</qs3/pubsys2/xml/en/procedures/377/377-011-070-tr.html>)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine to normal operating temperature. Check for leaks and proper operation.

If the malfunction resulted in coolant, oil, excessive fuel, or excessive black smoke entering the exhaust system, the aftertreatment system **must** be inspected.

- Use the following procedure to check the aftertreatment diesel particulate filter (DPF). Refer to Procedure 014-016 in Section 14. (</qs3/pubsys2/xml/en/procedures/493/493-014-016.html>)